Lecture-6

Tutorials on

1. Mass action kinetics and equilibrium
2. Michaelis-Menten kinetics
3. Hill's equation

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Writing rate laws, stoichiometry and evolution of concentration

- Consider the following reactions (i) $A + B \rightarrow C + D$, (ii) $3\ A + B \rightarrow A + C$ and (iii) $A + 2\ B \rightarrow 3A + c$.
  Assume k to be the rate constant of the all the reactions.
- What is the rate law, 'v' for the reaction?
- What are the stoichiometric coefficients for the reaction?
- What is the integrated rate law (or) evolution of the concentration for each of the reactions?

Solution-(i)

- The rate of irreversible reaction is given as $v =$ k [A][B].
- The evolution equations or the concentration variations in time for the reactant [A], [B], and the products [C] and [D] are given as $\frac{d[A]}{dt}$, $\frac{d[B]}{dt}$, $\frac{d[C]}{dt}$, $\frac{d[D]}{dt}$.
- Stoichiometric coefficients for A, B, C, D are -1, -1, 1, and 1 respectively.
- The evolution /integrated rate laws of the species depends on the rate $\frac{1}{\eta_A}\frac{d[A]}{dt} = \frac{1}{\eta_B}\frac{d[B]}{dt} = v; -\frac{d[A]}{dt} = -\frac{d[B]}{dt} = v$
  $\frac{1}{\eta_C}\frac{d[C]}{dt} = \frac{d[D]}{dt} = v; -\frac{d[C]}{dt} = \frac{d[D]}{dt} = v$
- Note the signs; -^{ve} for reactant & $+^{ve}$ for product.

Solution-(ii)

- The rate of the reaction $3\ A + B \rightarrow A + C$ is given as $v = k[A]^3[B]$.
- When we write the evolution of the concentration of A, we must take into consideration the fact that each time this reaction occurs, only two molecules of A are transformed (one is conserved). So, the variation of A is given by $-\frac{1}{2}\frac{d[A]}{dt} = v = k\ [A]^3[B]$.
- The coefficient 2 is the balance for the species A in reaction and the sign – stands because, globally, A is transformed (not produced).
- Three things are vital to write a correct rate laws for a given reaction: (i) Sign, (ii) Stoichiometric coefficient, & (iii) Rate, v.

Solution - (iii)

- For the reaction A + 2 B $\rightarrow$ 3A + C, the stoichiometric coefficients are
  $\eta_A = 3 - 1 = 2$, $\eta_B = 0 - 2 =$ -2, $\eta_C = 1 - 0 = 1$.

- So $\frac{d[A]}{dt} = \eta_A\ k[\mathrm{A}][\mathrm{B}]^2 = 2k\ [\mathrm{A}][\mathrm{B}]^2$
  $-\frac{d[B]}{dt} = \eta_B k\ [\mathrm{A}][\mathrm{B}]^2 = 2k\ [\mathrm{A}][\mathrm{B}]^2$
  $\frac{d[C]}{dt} = \eta_C k[\mathrm{A}][\mathrm{B}]^2 = k[\mathrm{A}][\mathrm{B}]^2$

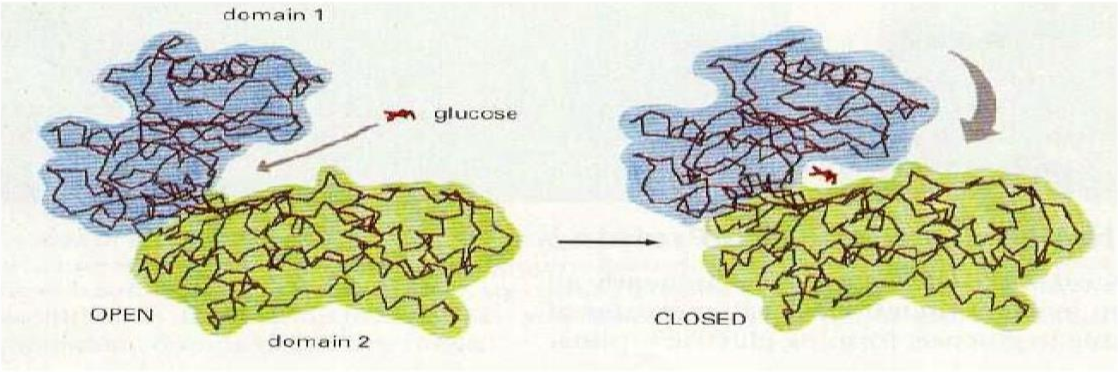
- **In general**, for the reaction n X + ... p X, the vectorial form of the equation is $\boxed{\frac{d[X]}{dt} = \eta v, \textit{with} \quad \eta = p - n}$.

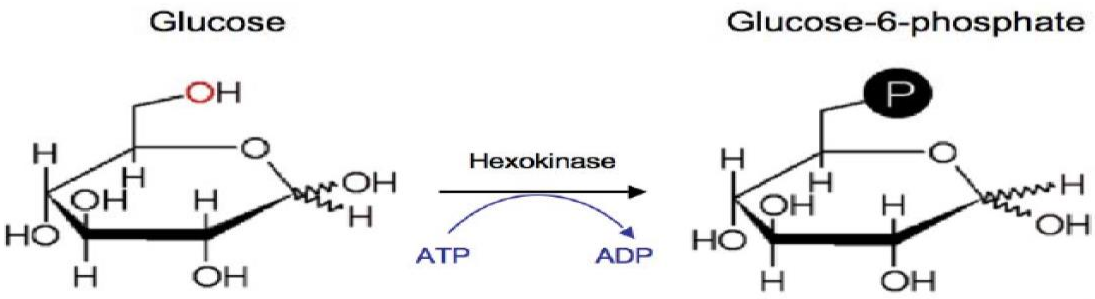
- η is called the stoichiometric coefficient. The coefficient is positive if, globally, the species is produced (p $>$ n), and negative if the species is transformed (n $>$ p).

Michealis-Menten kinetics-Example

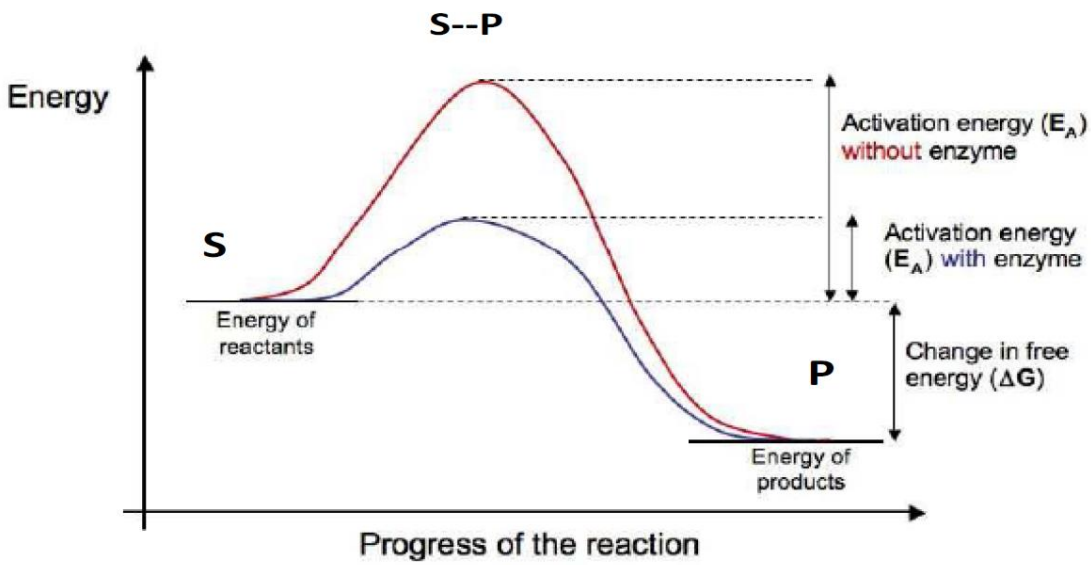
- An example of an enzymatic reaction is the first reaction of the glycolysis, catalysed by the enzyme hexokinase/

$$\text{Glucose} + \text{ATP} \rightarrow \text{Glucose-6-phosphate} + \text{ADP}$$





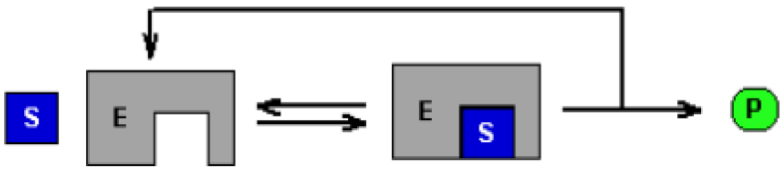
How enzyme catalyze the reaction?



- In the above reaction $S \underset{k_{-1}=10^{-5}}{\overset{k_1=10^{-3}}{\rightleftharpoons}} \mathrm{P} and k_{eq} = \frac{[P]}{[S]} = 100.$
- If both k_1 and k_{-1} increase by 10^6 fold by enzyme, still K_{eq} will be

$$k_{eq} = \frac{10^{(-3+6)}}{10^{(-5+6)}} = 100.$$

Enzyme catalysis - Derivation of Michealis-Menten equation



- The reaction scheme can be written as follows (C=complex between E and S, namely [ES]):
- $$E + S \underset{k_{-1}}{\overset{k_1}{\longleftrightarrow}} C \xrightarrow{k_2} E + P$$
- The equation for the different species follow the mass action and the evolution law is given as:

Derivation of Michealis-Menten equation

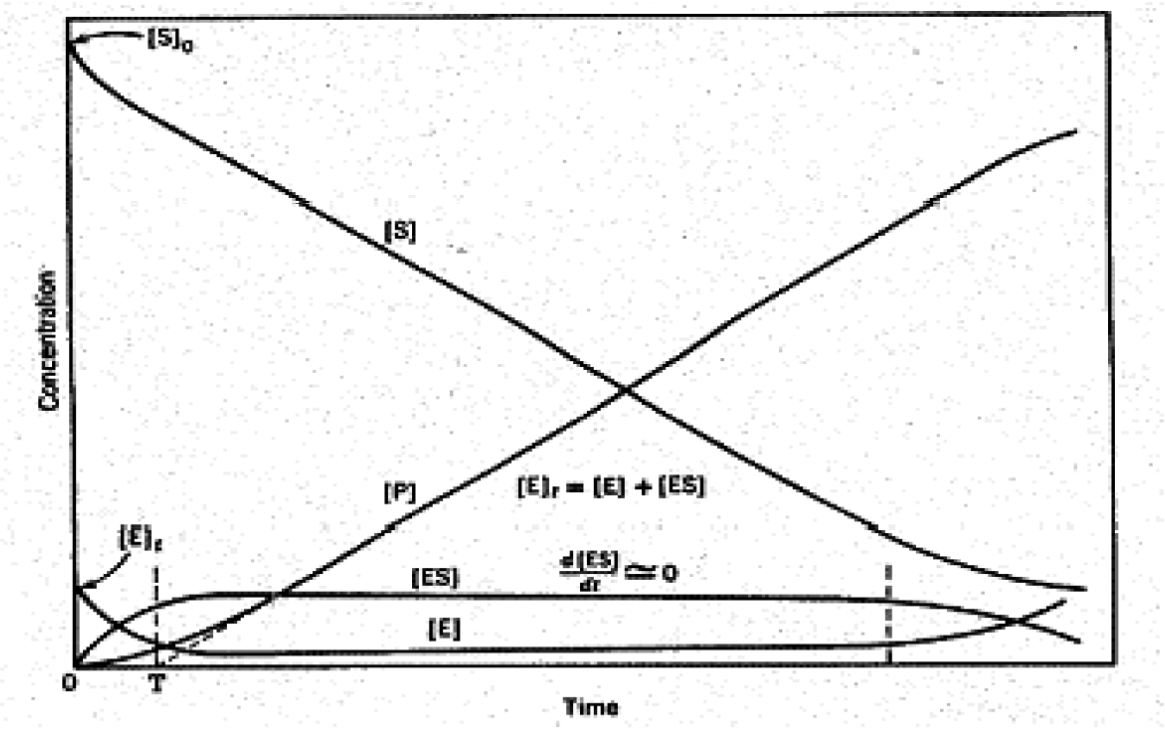
$$\frac{d[S]}{dt} = -k_1[E][S] + k_{-1}[C]$$

$$\frac{d[E]}{dt} = -k_1[E][S] + k_{-1}[C] + k_2[C]$$

$$\frac{d[C]}{dt} = k_1[E][S] - k_{-1}C - k_2[C]$$

$$\frac{d[P]}{dt} = k_2[C]$$

Quasi-steady state approximation



- In this $\frac{d[C]}{dt} = \frac{d[E]}{dt} = 0$
- We define E_{tot} the total concentration of enzyme: $E_{tot} = E + C =$ constant

Quasi-steady state approximation

$$
\begin{aligned}
\frac{d[S]}{dt} &= -k_1[E][S] + k_{-1}[C] \\
\frac{d[E]}{dt} &= -k_1[E][S] + k_{-1}[C] + k_2[C] = 0 \\
\frac{d[C]}{dt} &= k_1[E][S] - k_{-1}C - k_2[C] = 0 \\
\frac{d[P]}{dt} &= k_2[C]
\end{aligned}
$$

- The rate of change of total enzyme and complex is

$$\boxed{\frac{d[E]}{dt} = 0, \frac{d[C]}{dt} = 0}$$

- From this equation, $E_{tot} = E + C$, we can extract C, and its given as

$$\boxed{C = \frac{k_1 E_{tot}[S]}{k_1[S] + (k_1 + k2)} = \frac{E_{tot}[S]}{[S] + \frac{(k_1+k2)}{k_1}}}$$

Final equation

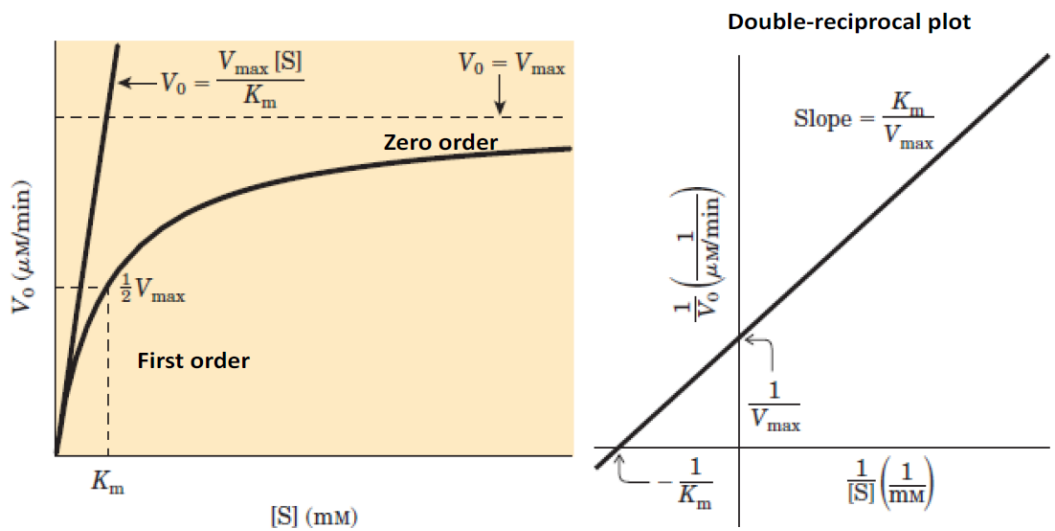
- When we replace this expression for C in the rate of appearance of P, we obtain:
  $v = \frac{d[P]}{dt} = \mathrm{k}_2[\mathrm{C}] = \frac{k_2 E_{tot}[S]}{[S]+\frac{(k_1+k2)}{k_1}}$
- This is usually written as $\boxed{v = v_{vmax} \frac{S}{S+K} M}$
- This parameters $\mathrm{K}_M = \frac{k_1+k_2}{k_1}$ and $\mathrm{v}_{max} = \mathrm{k}_2\ \mathrm{E}_{TOT}$.
- The rate is thus similar than in the case of the equilibrium hypothesis (Michaelis-Menten equation); only K_M has a slightly different meaning.
- We see that when K_1, $\mathrm{K}_{-1} >> \mathrm{K}_2$, we have $\mathrm{K}_M \rightarrow \mathrm{K}_S$. Note that K_M is usually called the Michaelis-Menten constant, although the exact meaning of this constant is rarely specified.

Lineweaver-Burk plot to estimate parameters

- Rewritten in the following manner, the equation gives a straight line, which is useful to determinate the parameters K_M and V_{max} (Lineweaver-Burk representation):

$$\frac{1}{v} = \frac{1}{v_{max}} + \frac{K_M}{v_{max}} \frac{1}{S}$$

- Can the two order of MM reactions be determined from elasticity ?



Steps involved in deriving rate laws in MM kinetics involving steady states

- **Step-1:** Identify the product and write the rate(s). For example, in MM kinetics, the reaction $E + S \underset{k_{-1}}{\overset{k_1}{\leftrightarrow}} C \xrightarrow{k_2} E + P$
- Its a two step reaction. The rate of product formation is $v = k_2\ [EX]$.
- **Step-2:** Write the enzyme conservation relationship. In the above case it is $E_T = [E] + [EX]$
- **Step-3:** Write the $\frac{numerator}{denominator}$ in the form of $\frac{v}{E_T}$
- **Step-4:** All the enzyme conservation terms in Step-2 should be written in terms of free enzymes. For this complexes are written in terms of free enzyme by making steady state approximations.

Continued...

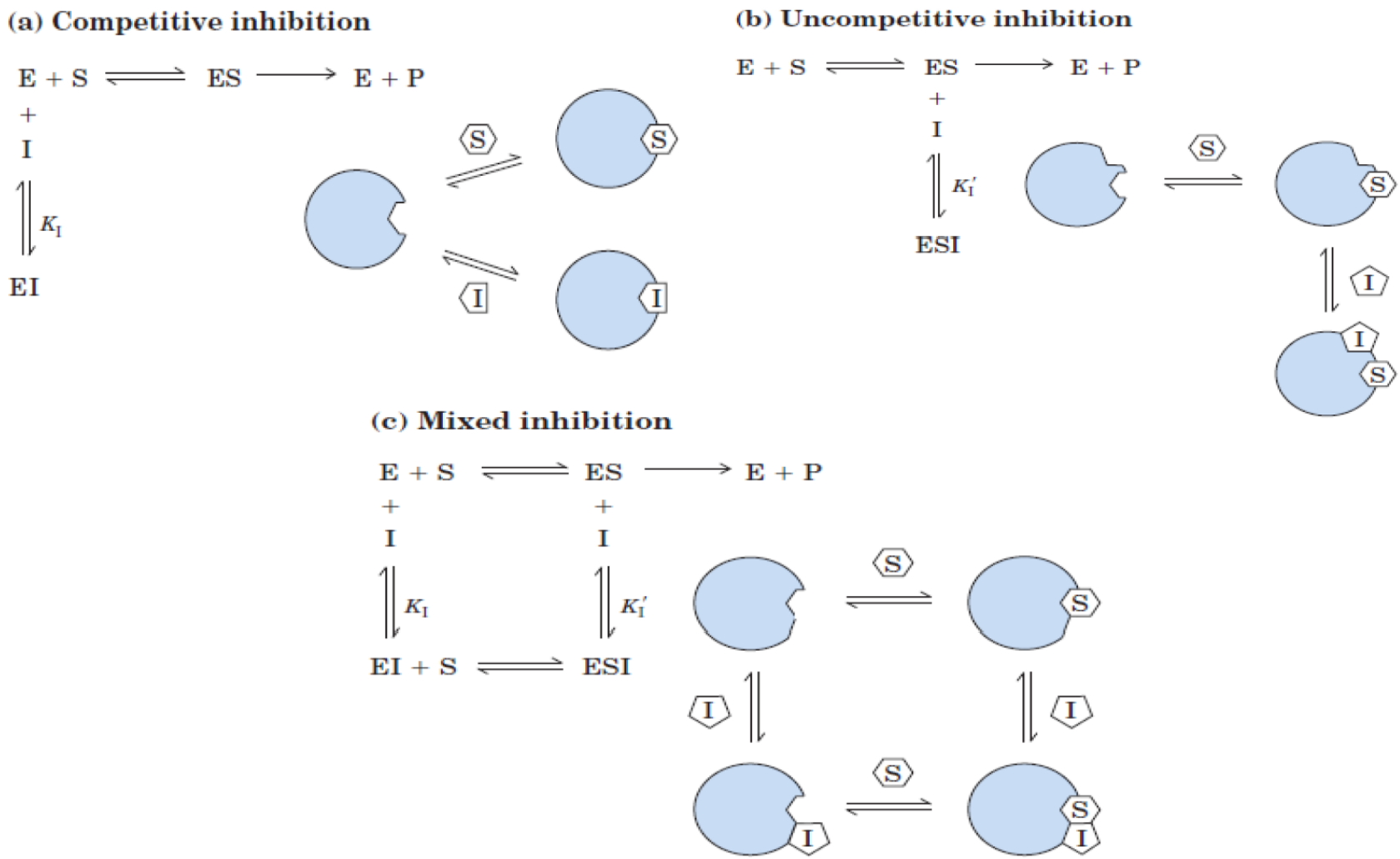
- For example, [EX] at steady state is written as k_1 [E][X] = k_2 [EX]. And [EX] = $\frac{k_1[E][X]}{k_2}$ = [EX] = $\frac{[E][X]}{k_m}$, where $\frac{k_2}{k_1} = k_m$.

- **Step-5:** Find $\frac{V}{E_T} = \frac{k_2[EX]}{[E]+[EX]} = \frac{\frac{k_2[E][X]}{k_m}}{E+\frac{[E][X]}{k_m}}$

- **Note:** In the denominator, the free enzyme is obtained and its cancelled by the free enzyme in the numerator. So, $\frac{k_2\frac{[E][x]}{k_m}}{[E][1+frac[x]k_m]} = \frac{k_2[x]}{k_m[1+\frac{[x]}{k_m}]}$.

- This leads finally to $\frac{v}{E_T} = \frac{k_2[x]}{k_m+[x]}$. Therefore v = $\frac{k_2 E_T[x]}{k_m+[x]}$ and if we write k_2 $E_T = v_m$, the final equation is $\boxed{v = \frac{v_m[x]}{k_m + [x]}}$

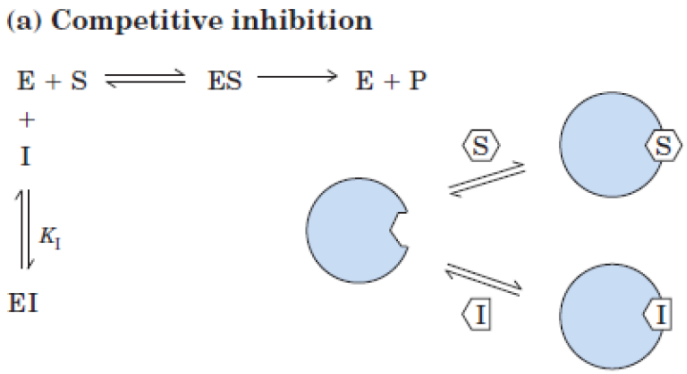
Enzyme inhibition{competitive and non-competitive

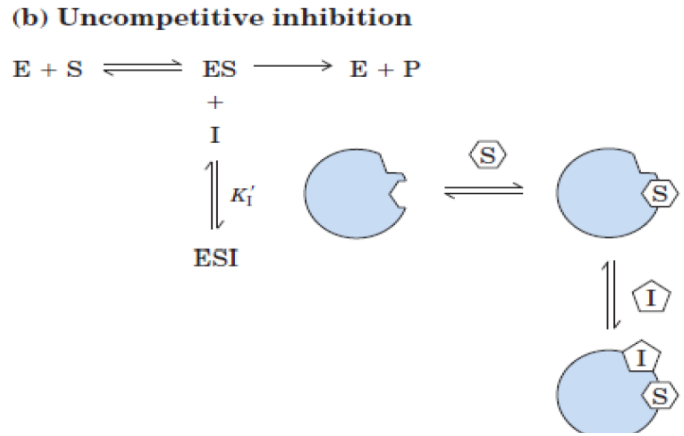


Steps involved in deriving competitive inhibition

- Write the chemical equation.
- Get E_{tot} and perform the SS approximation and Get the rate.
- Find the elasticities.
- The solution is $\boxed{v = v_{max} \frac{S}{K_M(1+\frac{I}{K_I})+S}}$.
- **Note:** Here, I is the Inhibitor.

Lineweaver Burke plot for inhibitions





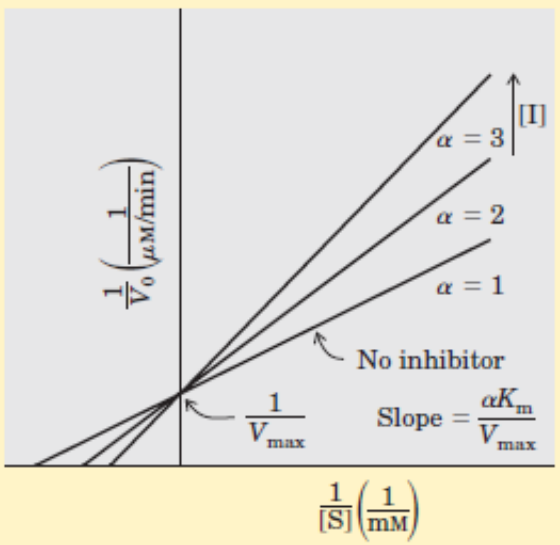


FIGURE 1 Competitive inhibition.

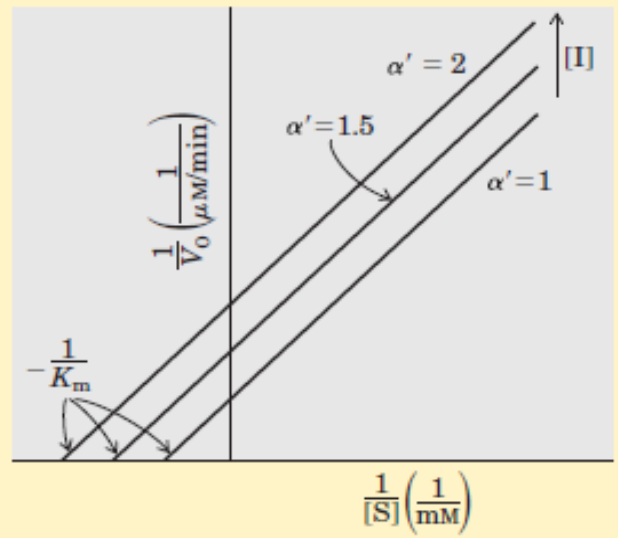
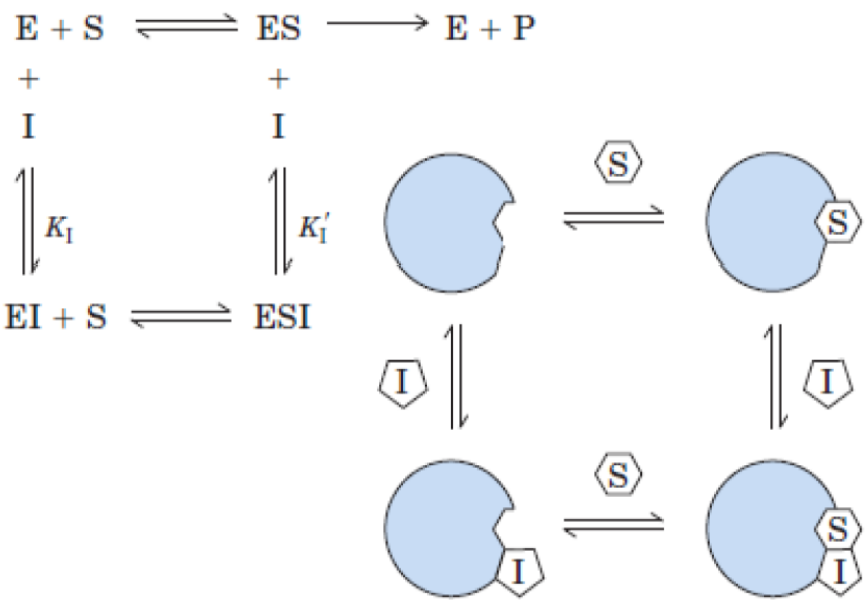


FIGURE 2 Uncompetitive inhibition.

Lineweaver Burke Plot for inhibition

(c) Mixed inhibition



$$\frac{1}{V_0} = \left(\frac{\alpha K_m}{V_{max}}\right)\frac{1}{[S]} + \frac{\alpha'}{V_{max}}$$

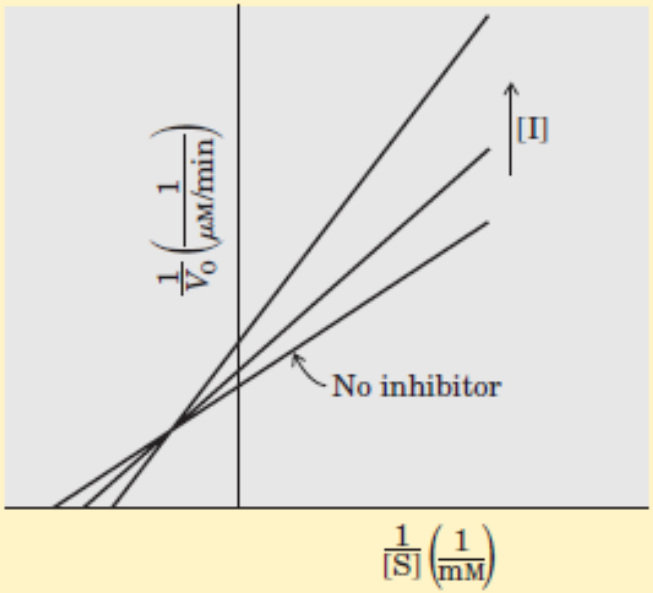


FIGURE 3 Mixed inhibition.

Hill's equation--Sigmoidal kinetics

To derive Hill's equation start with

$$E + nS \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES_n \xrightarrow{k_2} E + nP$$

and assume quasi-steady state approximation, $[ESn] = [E][S]^n/K^2_n$ where K^2_n is the association constant.Note that here $K^2_n = (k_1 + k_{-1} / k_2)$

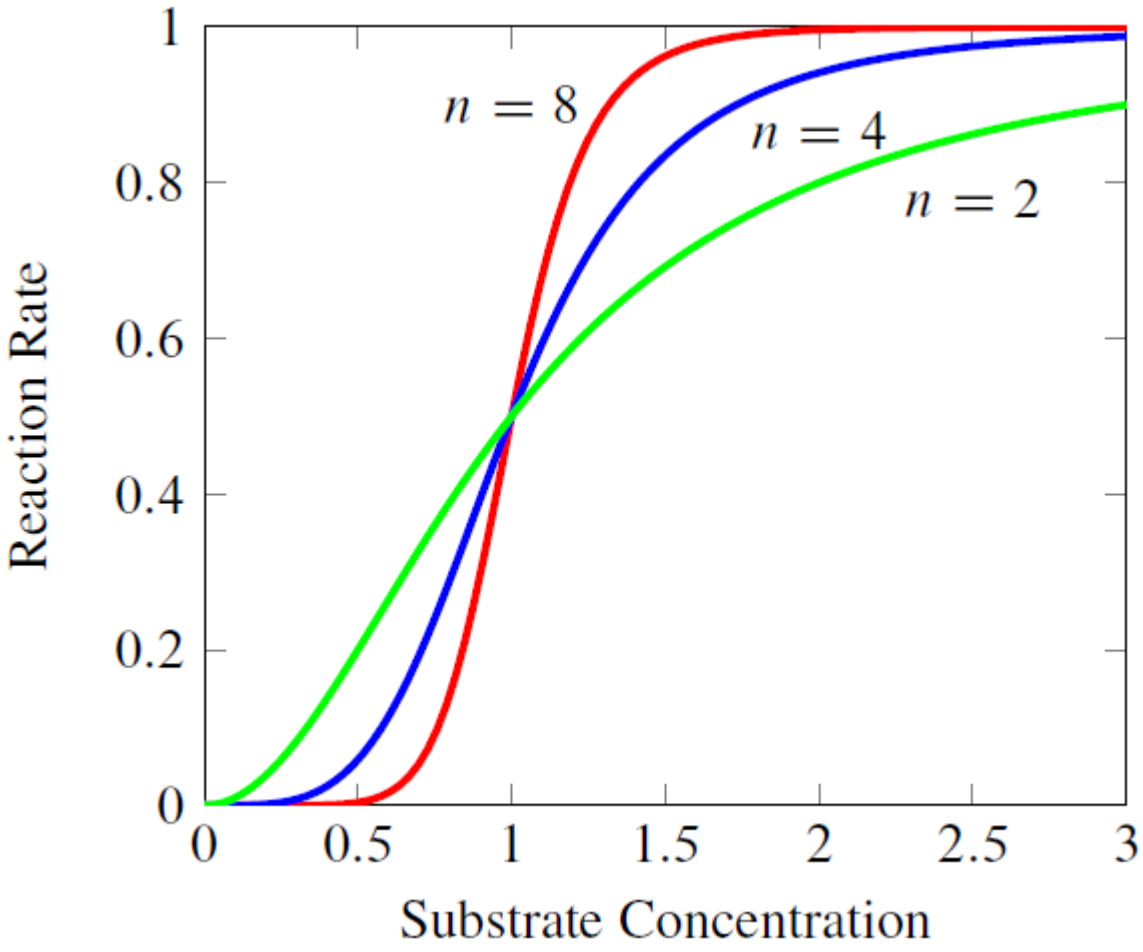
Using conservation relation, $E_T = E + ES_n$, the rate is given by then

$$E_T = E + [E][S]^n/K^2_n \quad = E\,(1 + [S]^n/ K^2_n)$$

$$v/ E_T \quad = k_2\,[ES_n] = k_2\,([E][S]^n/K^2_n)/\ E\,(1 + [S]^n/ K^2_n)$$

$$= k_2\,[S]^n/\ (K^2_n + [S]^n)$$

$$v = E_T\,k_2\,[S]^n/\ (K^2_n + [S]^n)$$

$$\boxed{v = v_{max}[S]^n/\ (K^2_n + [S]^n)}$$

Velocity curves for Hill's equation